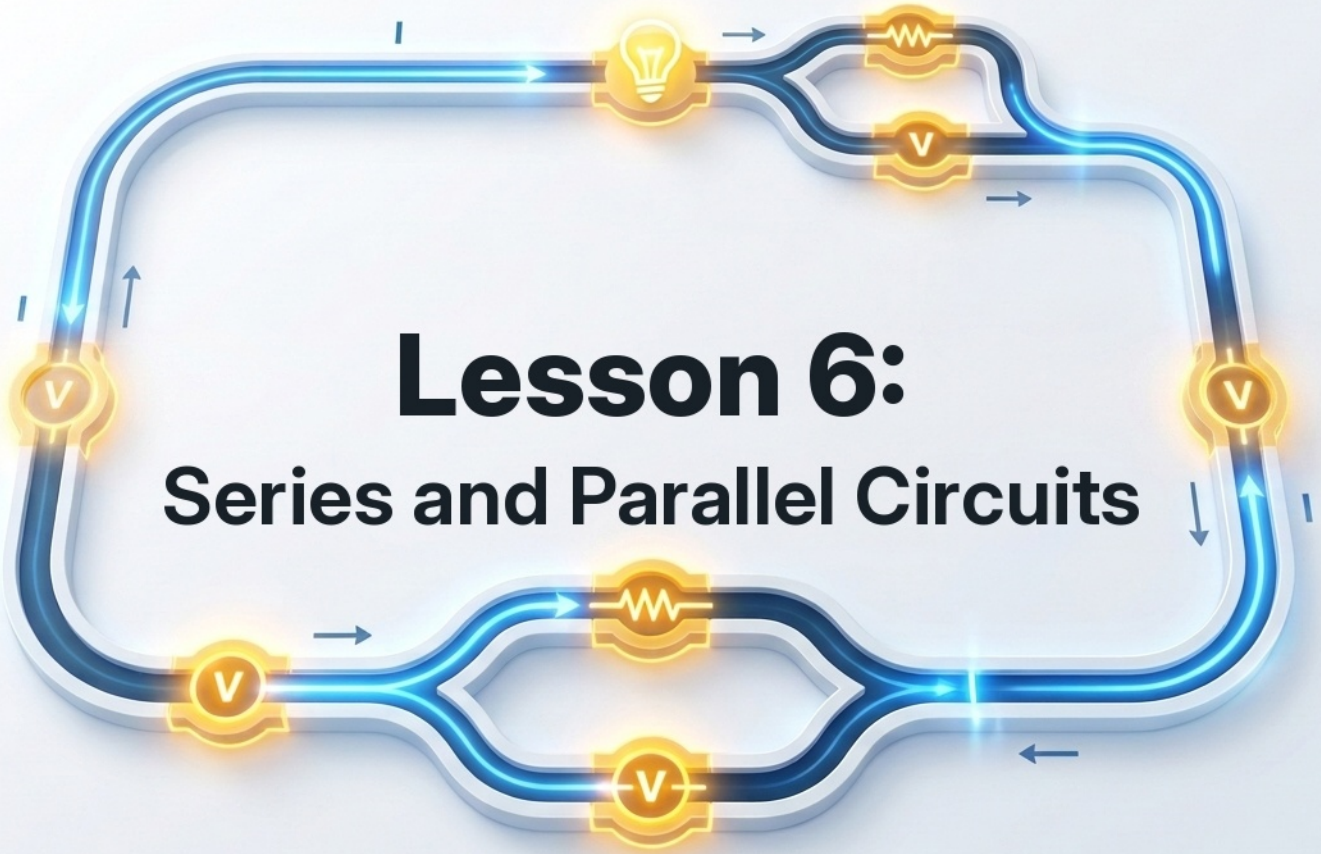


Lesson 6:

Series and Parallel Circuits

Gemini Notebook

In a series circuit, how does current compare at different points?

Success Cue: Only one path

Answer: Current is identical everywhere

In a parallel circuit, what is true about the voltage across each branch?

Success Cue: Full supply voltage

Answer: Voltage is equal across each parallel branch

If two 10 Ω resistors are connected in series, what is the total resistance?

Success Cue: $R_{total} = R_1 + R_2$

Answer: 20 Ω (10 + 10)

What happens to total resistance when a resistor is added in parallel?

Success Cue: More paths for current

Answer: Total resistance decreases

Why are domestic household lights wired in parallel rather than series?

Success Cue: Home wiring advantages

Answer: Can be switched independently and if one blows, others stay on

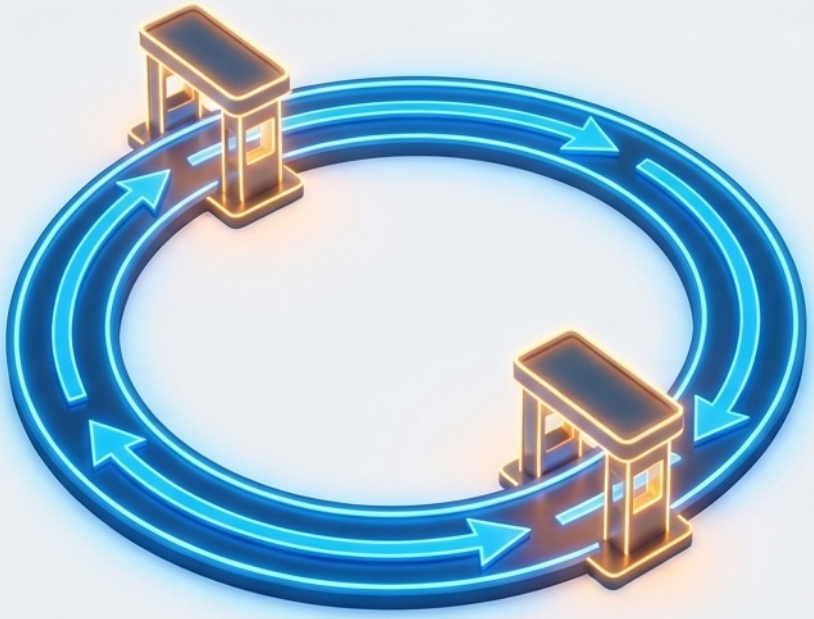
In a series circuit with a 12 V battery and two identical bulbs, what is the voltage across each?

Success Cue: V_{total} *shared*

Answer: 6 V (shared equally: 12 / 2)

Gemini Notebook

Series Circuits: The Single Loop



One Pathway: Current has only one route to travel.

Identical Current: Current (I) is exactly the same at all points.

Shared Voltage: Total potential difference (V) is split across components.

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Practical 1: Proving Series Rules

Equipment Checklist



Power supply (12V)



Ammeters and Voltmeters



2 and 3 filament lamps



Connecting leads

Goal

Prove $V_{total} = V_1 + V_2$ and **current is constant**.

Safety

Hot components: Lamps heat up quickly. Switch off between readings.

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Practical 1: Workflow & Results

Method Steps

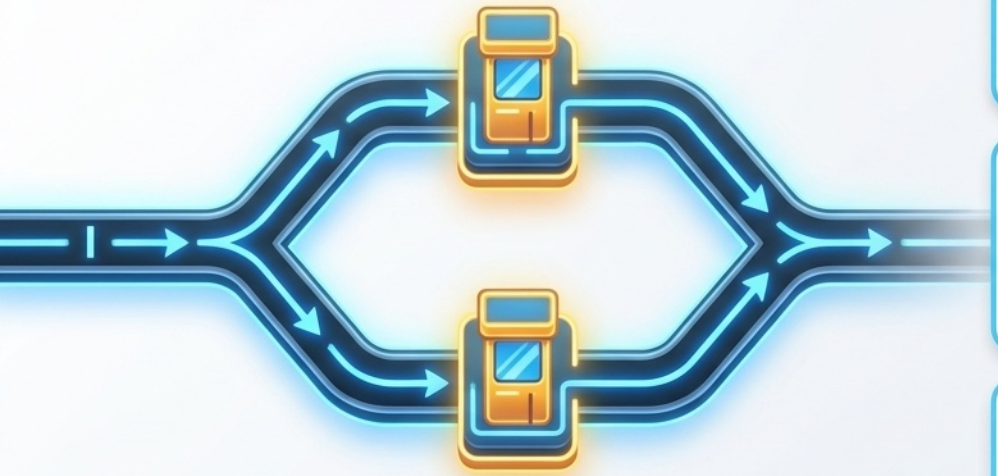
1. **Build:** Construct a single-loop series circuit with 2 lamps.
2. **Measure Current:** Place ammeter before, between, and after lamps.
3. **Measure Voltage:** Place voltmeter in parallel across each lamp and supply.

Position	Current (A)	Voltage (V)
Supply		
Lamp 1		
Lamp 2		
Lamp 3 (for extension)		

Does $V_1 + V_2 = V_{supply}$?

Gemini Notebook

Parallel Circuits: The Branching Network



Multiple Pathways: The circuit branches into separate, independent loops.

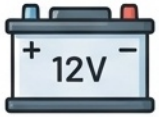
Split Current: Total current (I) divides across the branches.

Equal Voltage: Every parallel branch gets the full supply potential difference (V).

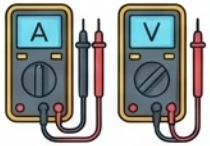
Gemini Notebook

Practical 2: Proving Parallel Rules

Equipment Checklist Box



- Power supply (12V)



- Ammeters and Voltmeters



- Resistors / Lamps



- Connecting leads

Goal:

Prove branch currents add up to supply current ($I_{total} = I_1 + I_2$) and parallel resistance drops.



Pro-Tip:

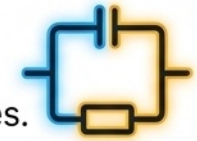
Circuit stacking: Build one complete loop first, then plug the parallel branch over the top of it.

Gemini Notebook

Practical 2: Workflow & Results

Method Steps

1 **Build:**
Construct a parallel circuit with 2 branches.



2 **Measure Current:**
Record supply current, then current in Branch 1 and Branch 2.



3 **Measure Voltage:**
Connect voltmeter across each parallel branch.



Component	Component	Current (A)	Voltage (V)
Main Supply			
Branch 1			
Branch 2			

Does $I_{Branch\ 1} + I_{Branch\ 2} = I_{supply}$?

Gemini Notebook

Circuit Rules: The Comparison Matrix

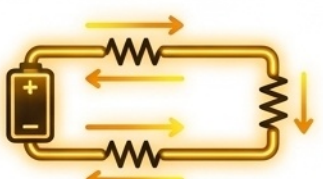
	Series	Parallel
I	$I_{total} = I_1 = I_2$ Same everywhere	$I_{total} = I_1 + I_2$ Splits at junctions
V	$V_{total} = V_1 + V_2$ Shared across components	$V_{total} = V_1 = V_2$ Equal across branches
R	$R_{total} = R_1 + R_2$ Adds up	Total R is less than the smallest resistor

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Checkpoint: Series Calculation Race

Setup: A 9V battery is connected to two resistors in series. Resistor A **9V** uses **4V**.

Question: What is the potential difference across Resistor B?



Expected Answer: 5V (because 9V - 4V = 5V)

Setup: Three 5 Ω lamps are wired in a single loop.

5 Ω

Question: What is the total circuit resistance?



Expected Answer: 15 Ω ($R_{total} = R_1 + R_2 + R_3$)

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The Resistance Mystery

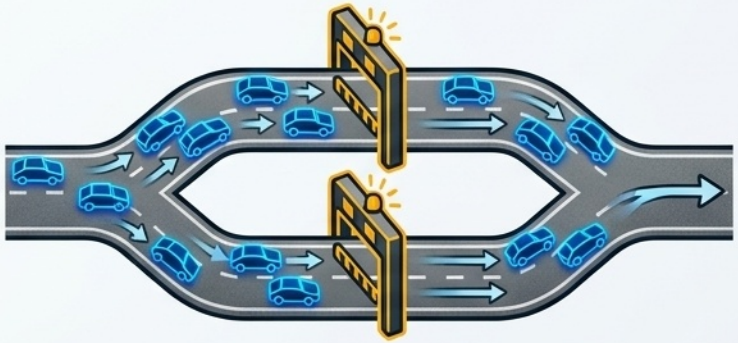
Series Circuit Metaphor



Series: Resistors act like consecutive tollgates. More tollgates = more traffic jams.

Total resistance **increases.**

Parallel Circuit Metaphor



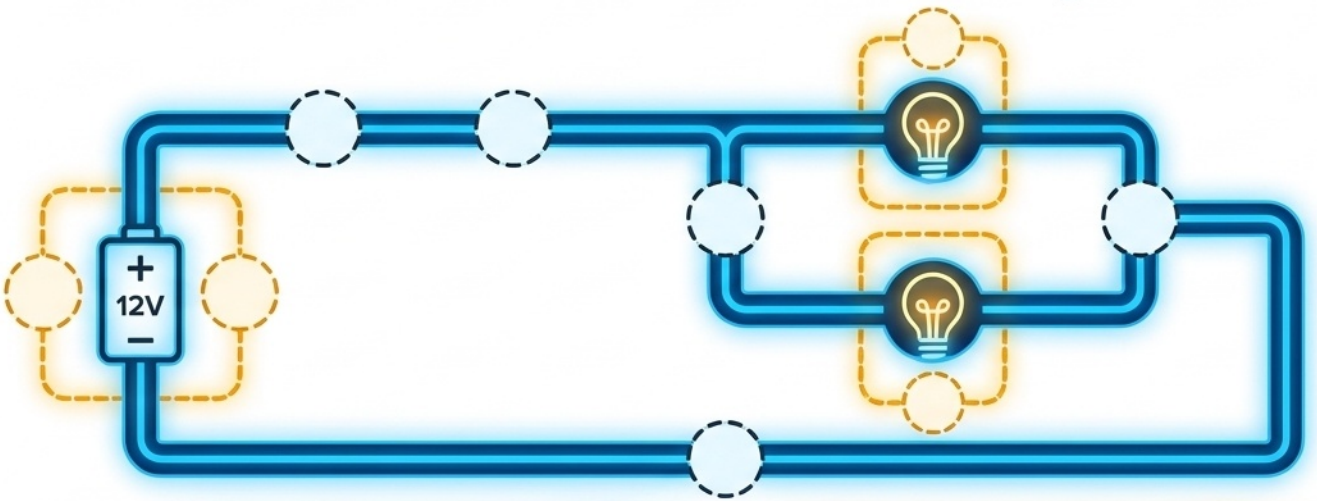
Parallel: Adding a branch opens a new road. More pathways = easier flow.

Total resistance **decreases.**

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Checkpoint: Diagram Labelling

Where do the Ammeters and Voltmeters go?





Ammeters go in **SERIES** (in the main traffic flow).

Voltmeters go in **PARALLEL** (hugging the component).

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Why are domestic household lights wired in parallel rather than series?



Hint 1

Hint 1: Think about what happens if a bulb breaks.

Hint 2

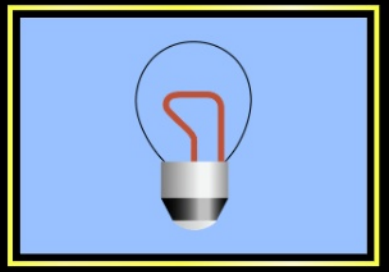
Hint 2: Think about the switches in your bedroom vs the kitchen.

Reveal

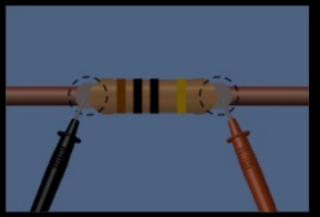
They can be switched independently, and if one blows, the rest stay on.

Gemini Notebook

Circuit Construction Kit: DC



Intro



Lab

Checkpoint: Concept Match

Series Circuit

Parallel Circuit

Adding components lowers total resistance.

If one component breaks, the whole circuit dies.

Voltage is shared across components.

Current is split across different branches.

Current is exactly the same everywhere.

Voltage is equal across all branches.

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Summary: The Core Rules

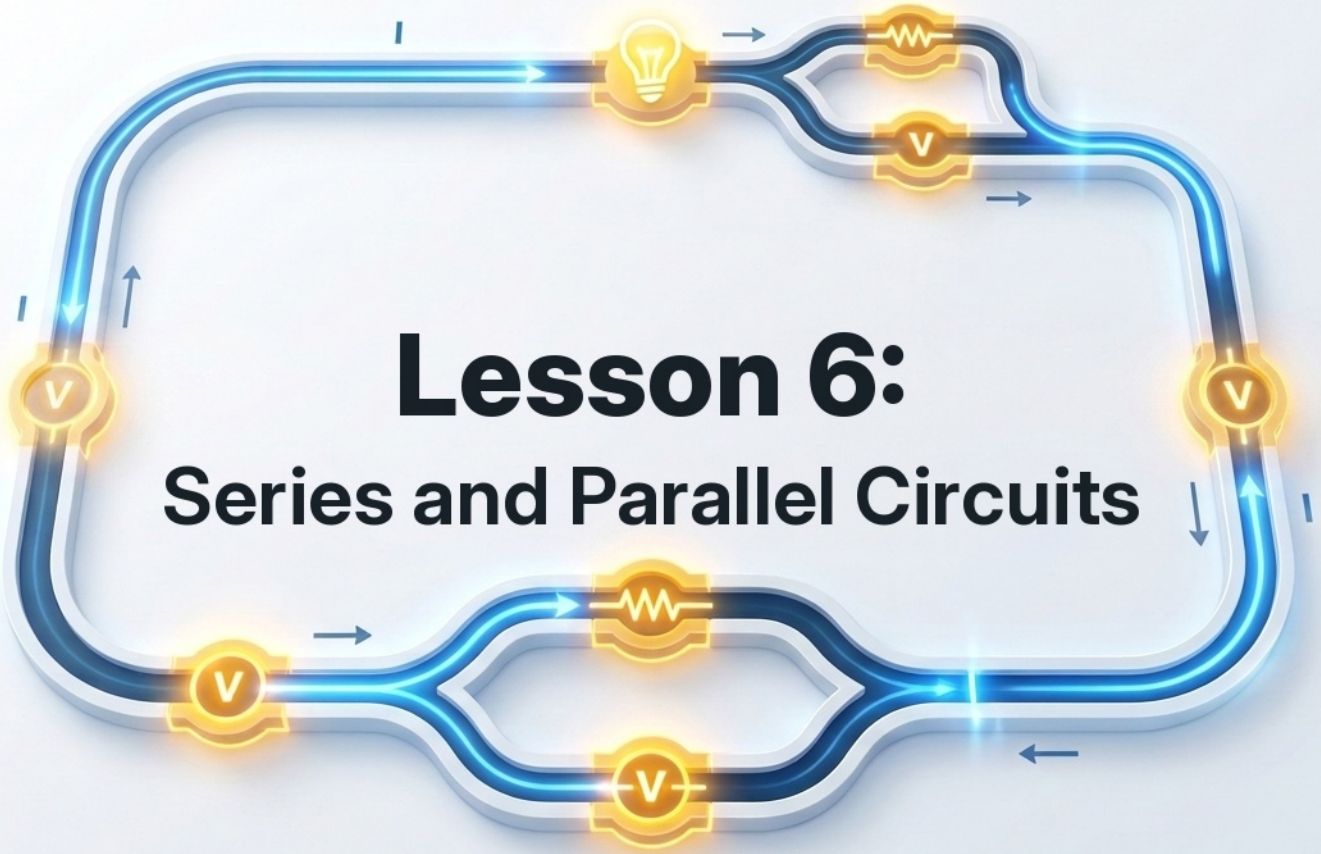
Series Circuits

- **One Path:** Current is the same everywhere.
- **Voltage Shared:** $V_{total} = V_1 + V_2$
- **Resistance Adds:** $R_{total} = R_1 + R_2$

Parallel Circuits

- **Branching Paths:** Current splits at junctions.
- **Voltage Equal:** Each branch gets full Voltage.
- **Resistance Drops:** Total R is less than the smallest branch.

Gemini Notebook



Lesson 6:

Series and Parallel Circuits

Gemini Notebook